

Лабораторная работа № 8.

Адресация IPv4 и IPv6. Настройка маршрутизации

Садова Д. А.

Российский университет дружбы народов, Москва, Россия

Информация

- Садова Диана Алексеевна
- студент бакалавриата
- Российский университет дружбы народов
- [113229118@pfur.ru]
- <https://DianaSadova.github.io/ru/>

Вводная часть

- В связи с исчерпанием адресного пространства IPv4 многие организации и интернет-провайдеры активно внедряют IPv6. Знание принципов работы IPv6, включая маршрутизацию и туннелирование, становится обязательным требованием для сетевых инженеров.

- Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

- Текст лабораторной работы № 8
- Интернет для исправления ошибок

Настройка динамической маршрутизации в сетях IPv4 и IPv6

Постановка задачи

Задана сеть, адреса сегментов сети, адреса, назначаемые интерфейсам устройств. Требуется настроить динамическую маршрутизацию по протоколам RIP, OSPF.

Топология моделируемой сети.

1. Запустите GNS3 VM и GNS3. Создайте новый проект.

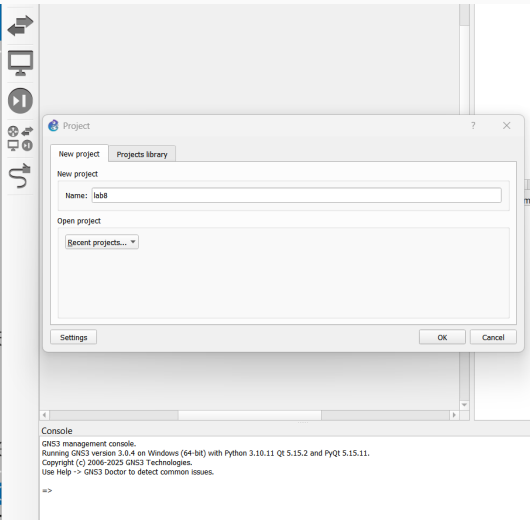
```
GNS3 server version: 3.0.4
Release channel: 3.0
VM version: 0.16.0
Ubuntu version: noble
Qemu version: 8.2.2
Virtualization: virtualbox
kvm
KVM support available: False
Uptime: up 0 minutes

IP: 192.168.56.105 PORT: 80

To log in using SSH: ssh gns3@192.168.56.105
Password: gns3

To launch the Web-Ui: http://192.168.56.105

Images and projects are stored in '/opt/gns3'
```



2. В рабочем пространстве разместите и соедините устройства в соответствии с топологией, приведённой на рис. 8.4. Используйте маршрутизаторы FRR.

3. Измените отображаемые названия устройств. Коммутаторам присвойте названия по принципу `msk-user-sw-0x`, маршрутизаторам — по принципу `msk-user-gw-0x`, VPCS — по принципу `PCx-user`, где вместо `user` укажите имя вашей учётной записи, вместо `x` — порядковый номер устройства.

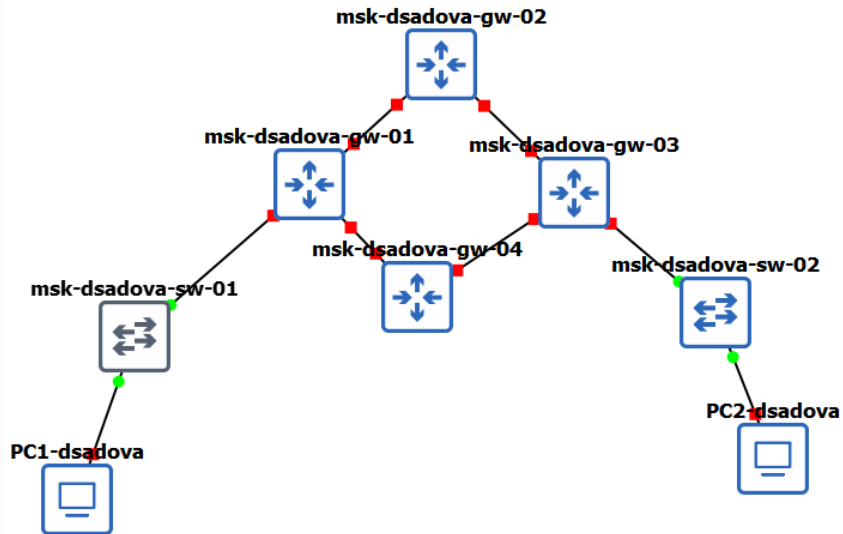


Рис. 2: Настройка имён устройств в GNS3

4. Включите захват трафика на соединении между коммутатором sw-01 и маршрутизатором gw-01, а также между коммутатором sw-02 и маршрутизатором gw-03.
5. Присвойте IPv4-адреса конечным устройствам PC1 и PC2 в соответствии с данными в табл. 8.2:

```
PC1-dsadova> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...

PC1-dsadova : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

PC1-dsadova>
PC1-dsadova> save
Saving startup configuration to startup.vpc
. done

PC1-dsadova> show ip

NAME           : PC1-dsadova[1]
IP/MASK         : 10.0.10.10/24
GATEWAY         : 10.0.10.1
DNS             :
MAC             : 00:50:79:66:68:00
LPORT          : 10066
RHOST:PORT      : 127.0.0.1:10067
MTU             : 1500

PC1-dsadova> 
```

Рис. 3: Настройка IPv4 на PC1

```
PC2-dsadova> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
PC2-dsadova : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC2-dsadova> save
Saving startup configuration to startup.vpc
. done

PC2-dsadova> show ip

NAME           : PC2-dsadova[1]
IP/MASK         : 10.0.11.10/24
GATEWAY         : 10.0.11.1
DNS             :
MAC             : 00:50:79:66:68:01
LPORT          : 10064
RHOST:PORT      : 127.0.0.1:10065
MTU             : 1500

PC2-dsadova> 
```

Рис. 4: Настройка IPv4 на PC2

6. Настройте IPv4-адреса на интерфейсах маршрутизаторов:

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-01
msk-dsadova-gw-01(config)# interface eth0
msk-dsadova-gw-01(config-if)# ip address 10.0.10.1/24
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth1
msk-dsadova-gw-01(config-if)# ip address 10.0.1.1/24
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth2
msk-dsadova-gw-01(config-if)# ip address 10.0.4.2/24
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# exit
msk-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-01#
```

```
msh-dsadova-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msh-dsadova-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msh-dsadova-gw-01#
```

Рис. 6: Настройка IPv4

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-02
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# ip address 10.0.1.2/24
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# interface eth1
msk-dsadova-gw-02(config-if)# ip address 10.0.2.1/24
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 7: Настройка IPv4

```
msk-dsadova-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
!
end
msk-dsadova-gw-02#
```

Рис. 8: Настройка IPv4

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-03
msk-dsadova-gw-03(config)# interface eth0
msk-dsadova-gw-03(config-if)# ip address 10.0.11.1/24
msk-dsadova-gw-03(config-if)# no shutdown
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# interface eth1
msk-dsadova-gw-03(config-if)# ip address 10.0.2.2/24
msk-dsadova-gw-03(config-if)# no shutdown
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# interface eth2
msk-dsadova-gw-03(config-if)# ip address 10.0.3.1/24
msk-dsadova-gw-03(config-if)# )# exit
% Unknown command: )# exit
msk-dsadova-gw-03(config-if)# msk-dsadova-gw-03(config)# i
% Unknown command: msk-dsadova-gw-03(config)# i
msk-dsadova-gw-03(config-if)# interface eth2
msk-dsadova-gw-03(config-if)# ip address 10.0.3.1/24
msk-dsadova-gw-03(config-if)# no shutdown
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# exit
msk-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-03#
```

Рис. 9: Настройка IPv4

```
msk-dsadova-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-dsadova-gw-03#
```

Рис. 10: Настройка IPv4

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-04
msk-dsadova-gw-04(config)# interface eth0
msk-dsadova-gw-04(config-if)# ip address 10.0.3.2/24
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# interface eth1
msk-dsadova-gw-04(config-if)# ip address 10.0.4.1/24
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 11: Настройка IPv4

```
msk-dsadova-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-dsadova-gw-04#
```

Рис. 12: Настройка IPv4

7. Присвойте IPv6-адреса оконечным устройствам PC1 и PC2 в соответствии с данными в табл. 8.2:

```
PC1-dsadova> ip 2001:10::a/64
PC1 : 2001:10::a/64

PC1-dsadova> save
Saving startup configuration to startup.vpc
. done

PC1-dsadova> show ipv6

NAME                : PC1-dsadova[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 10064
RHOST:PORT          : 127.0.0.1:10065
MTU                 : 1500
```

```
PC2-dsadova> ip 2001:11::a/64
PC1 : 2001:11::a/64

PC2-dsadova> save
Saving startup configuration to startup.vpc
. done

PC2-dsadova> show ipv6

NAME                : PC2-dsadova[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 10066
RHOST:PORT           : 127.0.0.1:10067
MTU                 : 1500

PC2-dsadova> █
```

Рис. 14: Настройка IPv6 на PC2

8. Настройте IPv6-адреса на интерфейсах маршрутизаторов:

```
msh-dsadova-gw-01# configure terminal
msh-dsadova-gw-01(config)# ipv6 forwarding
msh-dsadova-gw-01(config)# interface eth0
msh-dsadova-gw-01(config-if)# ipv6 address 2001:10::1/64
msh-dsadova-gw-01(config-if)# no ipv6 nd suppress-ra
msh-dsadova-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msh-dsadova-gw-01(config-if)# no shutdown
msh-dsadova-gw-01(config-if)# exit
msh-dsadova-gw-01(config)# interface eth1
msh-dsadova-gw-01(config-if)# ipv6 address 2001:1::1/64
msh-dsadova-gw-01(config-if)# no shutdown
msh-dsadova-gw-01(config-if)# exit
msh-dsadova-gw-01(config)# no ipv6 nd suppress-ra
% Unknown command: no ipv6 nd suppress-ra
va-gw-01(config-if)# msh-dsadova-gw-01(config)#
msh-dsadova-gw-01(config)# interface eth2
msh-dsadova-gw-01(config-if)# ipv6 address 2001:4::2/64
msh-dsadova-gw-01(config-if)# no shutdown
msh-dsadova-gw-01(config-if)# exit
msh-dsadova-gw-01(config)# exit
msh-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msh-dsadova-gw-01#
```

```
msk-dsadova-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-01
service integrated-vtysh-config
!
interface eth0
    ip address 10.0.10.1/24
    ipv6 address 2001:10::1/64
    ipv6 nd prefix 2001:10::/64
    no ipv6 nd suppress-ra
exit
!
interface eth1
    ip address 10.0.1.1/24
    ipv6 address 2001:1::1/64
exit
!
interface eth2
    ip address 10.0.4.2/24
    ipv6 address 2001:4::2/64
exit
!
end
msk-dsadova-gw-01#
```

Рис. 16: Настройка IPv6

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# ipv6 forwarding
msk-dsadova-gw-02(config)#
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# inal
    ipv6 f% Unknown command: inal
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# interface eth1
msk-dsadova-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02# █
```

Рис. 17: Настройка IPv6

```
msk-dsadova-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
exit
!
end
msk-dsadova-gw-02#
```

Рис. 18: Настройка IPv6

```
msh-dsadova-gw-03# configure terminal
msh-dsadova-gw-03(config)# ipv6 forwarding
msh-dsadova-gw-03(config)# interface eth0
msh-dsadova-gw-03(config-if)# ipv6 address 2001:11::1/64
msh-dsadova-gw-03(config-if)# no ipv6 nd suppress-ra
msh-dsadova-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msh-dsadova-gw-03(config-if)# no shutdown
msh-dsadova-gw-03(config-if)# exit
msh-dsadova-gw-03(config)# interface eth1
msh-dsadova-gw-03(config-if)# ipv6 address 2001:2::2/64
msh-dsadova-gw-03(config-if)# no shutdown
msh-dsadova-gw-03(config-if)# exit
msh-dsadova-gw-03(config)# interface eth2
msh-dsadova-gw-03(config-if)# ipv6 address 2001:3::1/64
msh-dsadova-gw-03(config-if)# no shutdown
msh-dsadova-gw-03(config-if)# exit
msh-dsadova-gw-03(config)# exit
msh-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msh-dsadova-gw-03#
```

Рис. 19: Настройка IPv6

```
msk-dsadova-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
interface eth2
 ip address 10.0.3.1/24
 ipv6 address 2001:3::1/64
exit
!
end
msk-dsadova-gw-03#
```

Рис. 20: Настройка IPv6


```
msk-dsadova-gw-04# configure terminal
msk-dsadova-gw-04(config)# ipv6 forwarding
msk-dsadova-gw-04(config)# interface eth0
msk-dsadova-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# interface eth1
msk-dsadova-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 21: Настройка IPv6

```
msk-dsadova-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-dsadova-gw-04
service integrated-vtysh-config
!
interface eth0
    ip address 10.0.3.2/24
    ipv6 address 2001:3::2/64
exit
!
interface eth1
    ip address 10.0.4.1/24
    ipv6 address 2001:4::1/64
exit
!
end
msk-dsadova-gw-04#
```

Рис. 22: Настройка IPv6

Настройка динамической маршрутизации по протоколу RIP.

1. На маршрутизаторах настройте RIP в качестве протокола динамической маршрутизации:

```
msk-dsadova-gw-01# configure terminal
msk-dsadova-gw-01(config)# router rip
msk-dsadova-gw-01(config-router)# version 2
msk-dsadova-gw-01(config-router)# network eth0
msk-dsadova-gw-01(config-router)# network eth1
msk-dsadova-gw-01(config-router)# network eth2
msk-dsadova-gw-01(config-router)# exit
msk-dsadova-gw-01(config)# exit
msk-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# router rip
msk-dsadova-gw-02(config-router)# version 2
msk-dsadova-gw-02(config-router)# network eth0
msk-dsadova-gw-02(config-router)# network eth1
msk-dsadova-gw-02(config-router)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 24: Настройка RIP на gw-02

```
msk-dsadova-gw-03# configure terminal
msk-dsadova-gw-03(config)# router rip
msk-dsadova-gw-03(config-router)# version 2
msk-dsadova-gw-03(config-router)# network eth0
msk-dsadova-gw-03(config-router)# network eth1
msk-dsadova-gw-03(config-router)# network eth2
msk-dsadova-gw-03(config-router)# exit
msk-dsadova-gw-03(config)# exit
msk-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-03#
```

Рис. 25: Настройка RIP на gw-03

```
msk-dsadova-gw-04# configure terminal
msk-dsadova-gw-04(config)# router rip
msk-dsadova-gw-04(config-router)# version 2
msk-dsadova-gw-04(config-router)# network eth0
msk-dsadova-gw-04(config-router)# network eth1
msk-dsadova-gw-04(config-router)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 26: Настройка RIP на gw-04

2. Убедитесь, что маршрутизация по RIP настроена:

```
msh-dsadova-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:02:40
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:00:31
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:01:11
msh-dsadova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface
```

	Network	Next Hop	Metric	From	Tag	Time
C(i)	10.0.1.0/24	0.0.0.0	1	self	0	
R(n)	10.0.2.0/24	10.0.1.2	2	10.0.1.2	0	02:35
R(n)	10.0.3.0/24	10.0.4.1	2	10.0.4.1	0	02:58
C(i)	10.0.4.0/24	0.0.0.0	1	self	0	
C(i)	10.0.10.0/24	0.0.0.0	1	self	0	
R(n)	10.0.11.0/24	10.0.1.2	3	10.0.1.2	0	02:35

```

msk-dsadova-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 16 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv   Key-chain
    eth0           2     2
    eth1           2     2
    eth2           2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway          BadPackets  BadRoutes   Distance  Last Update
    10.0.1.2          0           0           120       00:00:03
    10.0.4.1          0           0           120       00:00:06
  Distance: (default is 120)
msk-dsadova-gw-01# 

```

Рис. 28: Проверка RIP на gw-02

3. Проверьте пути прохождения пакетов: – С PC1 пропингуйте PC2 и определите путь следования пакетов:

```
PC1-dsadova> ping 10.0.11.10

10.0.11.10 icmp_seq=1 timeout
10.0.11.10 icmp_seq=2 timeout
10.0.11.10 icmp_seq=3 timeout
10.0.11.10 icmp_seq=4 timeout
10.0.11.10 icmp_seq=5 timeout

PC1-dsadova> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    1.972 ms  14.349 ms  4.015 ms
 2      * * *
 3  *10.0.10.1   123.557 ms (ICMP type:3, code:1, Destination host unreachable)

PC1-dsadova> █
```

Рис. 29: Traceroute с PC1 на PC2 через RIP

– Проверьте метрики протокола RIP:

```
msk-dsadova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:37
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:52
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.1.2          3 10.0.1.2           0 02:37
msk-dsadova-gw-01#
```

Рис. 30: Метрики RIP на gw-01

– Предположим, что пакет проходит через маршрутизатор msk-user-gw-02 (в противном случае для дальнейших действий используйте маршрутизатор msk-user-gw-04). Отключите на маршрутизаторе msk-user-gw-02 интерфейс

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# shutdown
```

Рис. 31: Отключение интерфейса на gw-02

– Проверьте метрики протокола RIP:

```
msk-dsadova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:37
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:52
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.1.2          3 10.0.1.2           0 02:37
msk-dsadova-gw-01#
```

Рис. 32: Метрики RIP после отключения интерфейса

– С PC1 пропикуйте PC2 и определите путь следования пакетов. Отметьте примерное время, за которое восстановится маршрут.

```
PC1-dsadova> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=26.374 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=14.689 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=25.907 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=14.059 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=29.817 ms

PC1-dsadova> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    20.836 ms   3.933 ms   1.972 ms
 2  10.0.1.2     15.842 ms   7.819 ms   5.217 ms
 3  10.0.2.2     18.992 ms   7.876 ms   3.884 ms
 4  10.0.11.10   30.509 ms   19.399 ms   9.888 ms
```

Рис. 33: Traceroute после отключения интерфейса

- Включите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02(config-if)# interface eth0  
msk-dsadova-gw-02(config-if)# no shutdown  
msk-dsadova-gw-02(config-if)#
```

Рис. 34: Включение интерфейса на gw-02

– С PC1 пропингуйте PC2 и определите путь следования пакетов.

```
msk-dsadova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:37
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:51
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.1.2          3 10.0.1.2           0 02:37
msk-dsadova-gw-01#
```

Рис. 35: Traceroute после восстановления интерфейса

– Посмотрите захваченный на соединениях трафик. В отчёте поясните, какую информацию можно извлечь из перехваченных пакетов.

163	1839.470944	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request id=0xffd4, seq=4/1024, ttl=64 (no response found!)
164	1841.469604	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request id=0x01d5, seq=5/1280, ttl=64 (no response found!)
165	1844.879906	10.0.10.10	10.0.11.10	TCP	74 37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765790980 TSecr=0

▶ Frame 163: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0	0000	0c 94 80 bd 00 00 00
▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)	0010	00 54 d4 fc 00 00 40
▶ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10	0020	0b 0a 08 00 20 33 ff
▼ Internet Control Message Protocol	0030	0e 0f 10 11 12 13 14
Type: Echo (ping) request (8)	0040	1e 1f 20 21 22 23 24
Code: 0	0050	2e 2f 30 31 32 33 34
Checksum: 0x2033 [correct]	0060	3e 3f
[Checksum Status: Good]		
Identifier (BE): 65492 (0xffd4)		
Identifier (LE): 54527 (0xd4ff)		
Sequence Number (BE): 4 (0x0004)		
Sequence Number (LE): 1024 (0x0400)		
▼ [No response seen]		
▶ [Expert Info (Warning/Sequence): No response seen to ICMP request]		
▶ Data (56 bytes)		

Рис. 36: Захват трафика

165	1844.879906	10.0.10.10	10.0.11.10	TCP	74	37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765790980 TSecr=0 WS=2
166	1844.881786	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
167	1844.883088	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=17657
168	1844.884946	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
169	1844.885634	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=17657
.....						
▶	Frame 165: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0					0000 0c 94 80 bd 00 00 00 50 79
▶	Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)					0010 00 3c 00 00 40 00 01 06 50
▶	Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10					0020 0b 0a 93 d1 93 d2 69 3f d5
▶	Transmission Control Protocol, Src Port: 37841, Dst Port: 37842, Seq: 0, Len: 0					0030 00 00 7d c7 00 00 02 04 05
						0040 d5 04 00 00 00 00 01 03 03

Рис. 37: Захват трафика

166	1844.881786	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
167	1844.883088	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=176575
168	1844.884946	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
169	1844.885634	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 37841 → 37842 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=176575

- Frame 166: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0 - Ethernet II, Src: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00) - Internet Protocol Version 4, Src: 10.0.10.1, Dst: 10.0.10.10 - Internet Control Message Protocol - Type: Time-to-live exceeded (11) [Expert Info (Note/Response): Type indicates an error] - Code: 0 (Time to live exceeded in transit) - Checksum: 0x1e42 [correct] [Checksum Status: Good] - Unused: 00000000 - Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10 - Transmission Control Protocol, Src Port: 37841, Dst Port: 37842, Seq: 1765790980	<pre> 0000 00 50 79 66 68 00 0c 94 80 b 0010 00 58 49 6b 00 00 40 01 08 7 0020 0a 0a 0b 00 1e 42 00 00 00 0 0030 40 00 01 06 50 a9 0a 00 0a 0 0040 93 d2 69 3f d5 04 00 00 00 0 0050 00 00 02 04 05 b4 01 01 08 0 0060 00 00 01 03 03 01 </pre>
--	--

Рис. 38: Захват трафика

→	49	1769.898592	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x6dd3, seq=5/1280, ttl=61 (reply in 50)	
←	50	1769.900203	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x6dd3, seq=5/1280, ttl=64 (request in 49)	
	51	1773.477323	10.0.10.10	10.0.11.10	TCP	74 38529 → 38530 [SYN]	Seq=0 Win=0 Len=0 MSS=1460 TSval=1765790577 TSecr=	
▶ Frame 49: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0								
▶ Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)								
▶ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10								
▶ Internet Control Message Protocol								
							0000	00 50 79 66 68 01
							0010	00 54 d3 6d 00 00
							0020	0b 0a 08 00 b2 33
							0030	0e 0f 10 11 12 13
							0040	1e 1f 20 21 22 23

Рис. 39: Захват трафика

```

51 1773.477323 10.0.10.10 10.0.11.10 TCP 74 38529 → 38530 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765790577 TSecr=0 WS=2
52 1773.477633 10.0.11.10 10.0.10.10 TCP 54 38530 → 38529 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
53 1773.517328 10.0.10.10 10.0.11.10 TCP 74 [TCP Retransmission] 38529 → 38530 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765790577 TSecr=0 WS=2
4
▶ Frame 51: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0
▼ Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Destination: Private_66:68:01 (00:50:79:66:68:01)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 9]
▼ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10
  0100 .... = Version: 4
  ....0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 60
  Identification: 0x0009 (9)
  010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 1
  Protocol: TCP (6)
  Header Checksum: 0x50a0 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.0.10.10
  Destination Address: 10.0.11.10
  [Stream index: 2]
▶ Transmission Control Protocol, Src Port: 38529, Dst Port: 38530, Seq: 0, Len: 0

```

Рис. 40: Захват трафика

- ICMP-запросы (ping) не получают ответа — маршрут отсутствует.

При отсутствии маршрута трафик не проходит дальше первого маршрутизатора, который возвращает ICMP-ошибки.

4. На маршрутизаторах настройте RIPng для сетей IPv6

```
msk-dsadova-gw-01# configure terminal
msk-dsadova-gw-01(config)# router ripng
msk-dsadova-gw-01(config-router)# network eth0
msk-dsadova-gw-01(config-router)# network eth1
msk-dsadova-gw-01(config-router)# network eth2
msk-dsadova-gw-01(config-router)# exit
msk-dsadova-gw-01(config)# exit
msk-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-01#
```

Рис. 41: Настройка RIPng

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# router ripng
msk-dsadova-gw-02(config-router)# network eth0
msk-dsadova-gw-02(config-router)# network eth1
msk-dsadova-gw-02(config-router)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02# █
```

Рис. 42: Настройка RIPng

```
msk-dsadova-gw-03# configure terminal
msk-dsadova-gw-03(config)# router ripng
msk-dsadova-gw-03(config-router)# network eth0
msk-dsadova-gw-03(config-router)# network eth1
msk-dsadova-gw-03(config-router)# network eth2
msk-dsadova-gw-03(config-router)# exit
msk-dsadova-gw-03(config)# exit
msk-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-03#
```

Рис. 43: Настройка RIPng


```
msk-dsadova-gw-04# configure terminal
msk-dsadova-gw-04(config)# router ripng
msk-dsadova-gw-04(config-router)# network eth0
msk-dsadova-gw-04(config-router)# network eth1
msk-dsadova-gw-04(config-router)# )
% Unknown command: )
msk-dsadova-gw-04(config-router)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 44: Настройка RIPng

5. Проверьте пути прохождения пакетов: – С PC1 пропингуйте PC2 и определите путь следования пакетов:

```
PC1-dsadova> ping 2001:11::a
```

```
2001:11::a icmp6_seq=1 ttl=58 time=28.866 ms
2001:11::a icmp6_seq=2 ttl=58 time=15.942 ms
2001:11::a icmp6_seq=3 ttl=58 time=15.192 ms
2001:11::a icmp6_seq=4 ttl=58 time=7.272 ms
2001:11::a icmp6_seq=5 ttl=58 time=10.217 ms
```

```
PC1-dsadova> trace 2001:11::a
```

```
trace to 2001:11::a, 64 hops max
```

1	2001:10::1	6.387 ms	4.107 ms	9.931 ms
2	2001:1::2	27.432 ms	15.614 ms	8.758 ms
3	2001:2::2	11.902 ms	11.351 ms	11.868 ms
4	2001:11::a	9.644 ms	11.682 ms	18.220 ms

– Проверьте метрики протокола RIPng:

```
msh-dsadova-gw-01# show ipv6 ripng
```

```
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
```

```
Sub-codes:
```

```
    (n) - normal, (s) - static, (d) - default, (r) - redistribute,
```

```
    (i) - interface, (a/S) - aggregated/Suppressed
```

	Network	Next Hop	Via	Metric	Tag	Time
C(i)	2001:1::/64					
		::	self	1	0	
R(n)	2001:2::/64					
		fe80::e74:9dff:febf:0	eth1	2	0	02:22
R(n)	2001:3::/64					
		fe80::e0c:4aff:febf:1	eth2	2	0	02:31
C(i)	2001:4::/64					
		::	self	1	0	
C(i)	2001:10::/64					
		::	self	1	0	
R(n)	2001:11::/64					
		fe80::e74:9dff:febf:0	eth1	3	0	02:31

```
msh-dsadova-gw-01#
```

– Предположим, что пакет проходит через маршрутизатор msk-user-gw-02 (в противном случае для дальнейших действий используйте маршрутизатор msk-user-gw-04). Отключите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 47: Отключение интерфейса на gw-02 (IPv6)

– Проверьте метрики протокола RIPng:

```
msk-dsadova-gw-01# show ipv6 ripng
```

```
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
```

```
Sub-codes:
```

```
    (n) - normal, (s) - static, (d) - default, (r) - redistribute,  
    (i) - interface, (a/S) - aggregated/Suppressed
```

	Network	Next Hop	Via	Metric	Tag	Time
C(i)	2001:1::/64					
		::	self	1	0	
R(n)	2001:2::/64					
		fe80::e74:9dff:fecf:0	eth1	2	0	02:01
R(n)	2001:3::/64					
		fe80::e0c:4aff:fef8:1	eth2	2	0	02:55
C(i)	2001:4::/64					
		::	self	1	0	
C(i)	2001:10::/64					
		::	self	1	0	
R(n)	2001:11::/64					
		fe80::e74:9dff:fecf:0	eth1	3	0	02:55

```
msk-dsadova-gw-01#
```

– С PC1 пропингуйте PC2 и определите путь следования пакетов.

```
PC1-dsadova> ping 2001:11::a

2001:11::a icmp6_seq=1 timeout
2001:11::a icmp6_seq=2 timeout
2001:11::a icmp6_seq=3 timeout
2001:11::a icmp6_seq=4 timeout
2001:11::a icmp6_seq=5 timeout

PC1-dsadova> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    11.280 ms  3.144 ms  4.691 ms
 2  * * *
 3 *2001:10::1  108.607 ms (ICMP type:1, code:3, Address unreachable)

PC1-dsadova> █
```

Рис. 49: Traceroute IPv6 после отключения интерфейса (попытка 1)

```

msk-dsadova-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

      Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64
      ::          self          1      0
R(n) 2001:2::/64
      fe80::e74:9dff:fecf:0    eth1      2      0  01:12
R(n) 2001:3::/64
      fe80::e0c:4aff:fef8:1    eth2      2      0  02:59
C(i) 2001:4::/64
      ::          self          1      0
C(i) 2001:10::/64
      ::          self          1      0
R(n) 2001:11::/64
      fe80::e74:9dff:fecf:0    eth1      3      0  02:59
msk-dsadova-gw-01# █

```

Рис. 50: Traceroute IPv6 после отключения интерфейса (попытка 2)

– Отметьте примерное время, за которое восстановится маршрут.

Время восстановления: $\approx 1-2$ секунды.

– Включите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 51: Включение интерфейса на gw-02 (IPv6)

Интерфейс успешно включён, сохранена конфигурация.

– С PC1 пропингуйте PC2 и определите путь следования пакетов

```
PC1-dsadova> ping 2001:11::a

*2001:10::1 icmp6_seq=1 ttl=64 time=0.000 ms (ICMP type:1, code:3, Address unreachable)
2001:11::a icmp6_seq=2 ttl=58 time=29.402 ms
2001:11::a icmp6_seq=3 ttl=58 time=26.891 ms
2001:11::a icmp6_seq=4 ttl=58 time=22.444 ms
2001:11::a icmp6_seq=5 ttl=58 time=19.398 ms

PC1-dsadova> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    1.421 ms   0.726 ms   0.594 ms
 2 2001:1::2     1.373 ms   6.181 ms   18.131 ms
 3 2001:2::2     10.381 ms  6.122 ms   6.484 ms
 4 2001:11::a    9.367 ms   5.005 ms   19.760 ms

PC1-dsadova> █
```

Traceroute показывает путь через три маршрутизатора 2001:10::1, 2001:1::2, 2001:2::2, 2001:11::a (PC2)

Всё работает корректно.

– Посмотрите захваченный на соединениях трафик. В отчёте поясните, какую информацию можно извлечь из перехваченных пакетов.

→	313	2485.736241	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x82d7, seq=4, hop limit=64 (reply in 314)		
←	314	2485.758064	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x82d7, seq=4, hop limit=58 (request in 313)		
1180001								
▶ Frame 313: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0							0000	0c 94 80 bd 00 00
▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0010	00 00 00 40 3a 40
▶ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a							0020	00 00 00 00 00 0a
▶ Internet Control Message Protocol v6							0030	00 00 00 00 00 0a

Рис. 53: Захват трафика IPv6

```
318 2489.134418 2001:10::a 2001:11::a UDP 126 22736 → 22737 Len=64
319 2489.135519 2001:10::1 2001:10::a ICMPv6 174 Time Exceeded (Hop limit exceeded in transit)
320 2489.136010 2001:10::a 2001:11::a UDP 126 22736 → 22737 Len=64
321 2489.136681 2001:10::1 2001:10::a ICMPv6 174 Time Exceeded (Hop limit exceeded in transit)
```

▶ Frame 319: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface -, id 0

- ▶ Ethernet II, Src: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
- ▶ Internet Protocol Version 6, Src: 2001:10::1, Dst: 2001:10::a
- ▼ Internet Control Message Protocol v6
 - ▼ Type: Time Exceeded (3)
 - ▶ [Expert Info (Note/Response): Type indicates an error]
 - Code: 0 (Hop limit exceeded in transit)
 - Checksum: 0x4b30 [correct]
 - [Checksum Status: Good]
 - Reserved: 00000000
 - ▶ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a
 - ▶ User Datagram Protocol, Src Port: 22736, Dst Port: 22737
 - ▶ Data (64 bytes)

Рис. 54: Захват трафика IPv6

```

→ 158 2817.245404 2001:10::a 2001:11::a ICMPv6 118 Echo (ping) request id=0x82d7, seq=4, hop limit=61 (reply in 159)
← 159 2817.246542 2001:11::a 2001:10::a ICMPv6 118 Echo (ping) reply id=0x82d7, seq=4, hop limit=61 (request in 158)

Frame 158: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Destination: Private_66:68:01 (00:50:79:66:68:01)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv6 (0x86dd)
  [Stream index: 9]
Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a
Internet Control Message Protocol v6

```

Рис. 55: Захват трафика IPv6

162	2820.697043	2001:10::a	2001:11::a	UDP	126 22736 → 22737 Len=64
163	2820.698609	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Mal
164	2820.704979	2001:10::a	2001:11::a	UDP	126 22736 → 22737 Len=64
165	2820.705229	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Mal

▶ Frame 162: Packet, 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0
 ▼ Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
 ▼ Destination: Private_66:68:01 (00:50:79:66:68:01)
 0. = LG bit: Globally unique address (factory default)
 0 = IG bit: Individual address (unicast)
 ▼ Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
 0. = LG bit: Globally unique address (factory default)
 0 = IG bit: Individual address (unicast)
 Type: IPv6 (0x86dd)
 [Stream index: 9]
 ▶ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a
 ▶ User Datagram Protocol, Src Port: 22736, Dst Port: 22737
 ▶ Data (64 bytes)

Рис. 56: Захват трафика IPv6

Настройка динамической маршрутизации по протоколу OSPF.

1. На маршрутизаторах настройте OSPFv2 для сетей IPv4:

```
msk-dsadova-gw-01# configure terminal
msk-dsadova-gw-01(config)# router ospf
msk-dsadova-gw-01(config-router)# network 10.0.10.0/24 area
% Command incomplete: network 10.0.10.0/24 area
msk-dsadova-gw-01(config-router)# network 10.0.10.0/24 area
% Command incomplete: network 10.0.10.0/24 area
msk-dsadova-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-dsadova-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-dsadova-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dsadova-gw-01(config-router)# exit
msk-dsadova-gw-01(config)# exit
msk-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```



```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# router ospf
msk-dsadova-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-dsadova-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-dsadova-gw-02(config-router)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 58: Настройка OSPFv2 на gw-02

```
msk-dsadova-gw-03# configure terminal
msk-dsadova-gw-03(config)# router ospf
msk-dsadova-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-dsadova-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-dsadova-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dsadova-gw-03(config-router)# exit
msk-dsadova-gw-03(config)# exit
msk-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-03#
```

Рис. 59: Настройка OSPFv2 на gw-03

```
msk-dsadova-gw-04# configure terminal
msk-dsadova-gw-04(config)# router ospf
msk-dsadova-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-dsadova-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-dsadova-gw-04(config-router)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 60: Настройка OSPFv2 на gw-04

2. С PC1 пропингуйте PC2 и определите путь следования пакетов:

```
PC1-dsadova> ping 10.0.11.10
```

```
84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=22.169 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=11.924 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=22.952 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=17.190 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=11.166 ms
```

```
PC1-dsadova> trace 10.0.11.10 -P 6
```

```
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
```

```
 1  10.0.10.1    22.008 ms  3.162 ms  3.288 ms
 2  10.0.4.1     15.553 ms  5.880 ms  3.253 ms
 3  10.0.3.1     10.590 ms  10.644 ms 9.497 ms
 4  10.0.11.10   14.938 ms  10.679 ms 32.515 ms
```

```
PC1-dsadova> 
```

Рис. 61: Traceroute через OSPFv2

3. Проверьте таблицу маршрутизации протокола OSPFv2:

```
msk-dsadova-gw-01# show ip ospf neighbor

Neighbor ID      Pri State           Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1          1 Full/Backup      2m35s             33.021s 10.0.1.2          et
h1:10.0.1.1
10.0.4.1          1 Full/Backup      46.156s           39.818s 10.0.4.1          et
h2:10.0.4.2
0 0 0
0 0 0

msk-dsadova-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
      directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
      via 10.0.1.2, eth1
N   10.0.3.0/24      [200] area: 0.0.0.0
      via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
      directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
      directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
      via 10.0.1.2, eth1
      via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-dsadova-gw-01#
```

4. Предположим, что пакет проходит через маршрутизатор msk-user-gw-02 (в противном случае для дальнейших действий используйте маршрутизатор msk-user-gw-04). Отключите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# shutdown
msk-dsadova-gw-02(config-if)#
```

Рис. 63: Отключение интерфейса на gw-02 (OSPF)

5. Проверьте таблицу маршрутизации протокола OSPFv2:

```
msk-dsadova-gw-01# show ip ospf neighbor

Neighbor ID      Pri State                Up Time           Dead Time Address      Int
erface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup          3m55s             12.453s 10.0.1.2      eth
1:10.0.1.1              1      0      0
10.0.4.1         1 Full/Backup          2m06s             39.261s 10.0.4.1      eth
2:10.0.4.2              0      0      0

msk-dsadova-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24          [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24          [300] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.3.0/24          [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24          [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24         [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24         [300] area: 0.0.0.0
    via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-dsadova-gw-01#
```

6. С PC1 пропикуйте PC2 и определите путь следования пакетов. Отметьте примерное время, за которое восстановится маршрут.

```
PC1-dsadova> ping 10.0.11.10
```

```
84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=31.823 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.042 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=13.035 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=10.559 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=11.802 ms
```

```
PC1-dsadova> trace 10.0.11.10 -P 6
```

```
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    8.361 ms  12.669 ms  6.749 ms
 2  10.0.4.1     7.256 ms  9.640 ms  10.644 ms
 3  10.0.3.1    12.950 ms  10.137 ms  5.488 ms
 4  10.0.11.10  11.321 ms  41.091 ms  22.190 ms
```


Ping успешен, среднее время ~14 мс.

Traceroute показывает альтернативный путь через 10.0.4.1, 10.0.3.1. Маршрут восстановился быстро

7. Включите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 66: Включение интерфейса на gw-02 (OSPF)

8. С PC1 пропингуйте PC2 и определите путь следования пакетов.

```
PC1-dsadova> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=18.648 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=12.488 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=20.307 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=19.725 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=7.197 ms

PC1-dsadova> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    2.490 ms  8.230 ms  5.070 ms
 2  10.0.1.2    43.692 ms 13.061 ms  6.263 ms
 3  10.0.2.2    27.104 ms  5.740 ms 12.903 ms
 4  10.0.11.10  10.260 ms  8.073 ms 10.889 ms

PC1-dsadova> 
```

Рис. 67: Traceroute после восстановления интерфейса в OSPF

Ping успешен, среднее время ~16 мс. Traceroute показывает путь через 10.0.1.2, 10.0.2.2 (первоначальный маршрут).

9. Посмотрите захваченный на соединениях трафик. В отчёте поясните, какую информацию можно извлечь из перехваченных пакетов.

```
→ 557 4461.222763 10.0.10.10 10.0.11.10 ICMP 98 Echo (ping) request id=0x3cdf, seq=4/1024, ttl=64 (reply in 558)
← 558 4461.233107 10.0.11.10 10.0.10.10 ICMP 98 Echo (ping) reply id=0x3cdf, seq=4/1024, ttl=61 (request in 557)

Frame 557: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)
  Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Source: Private_66:68:00 (00:50:79:66:68:00)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 8]
Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10
Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xe328 [correct]
  [Checksum Status: Good]
  Identifier (BE): 15583 (0x3cdf)
  Identifier (LE): 57148 (0xdf3c)
  Sequence Number (BE): 4 (0x0004)
  Sequence Number (LE): 1024 (0x0400)
  [Response frame: 558]
Data (56 bytes)
```

Рис. 68: Захват трафика OSPF (пакет 1)

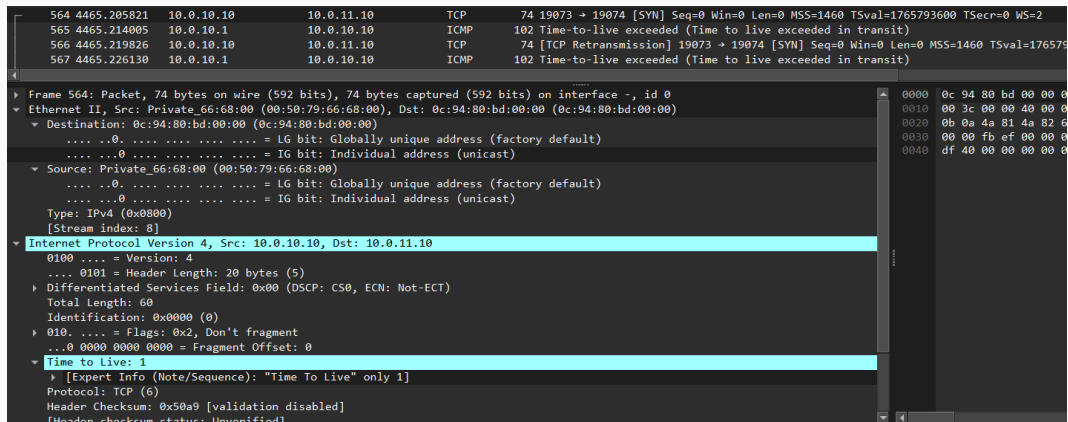


Рис. 69: Захват трафика OSPF (пакет 2)

```
→ 606 4551.755493 10.0.10.10 10.0.11.10 ICMP 98 Echo (ping) request id=0x97df, seq=3/768, ttl=64 (reply in 607)
← 607 4551.774775 10.0.11.10 10.0.10.10 ICMP 98 Echo (ping) reply id=0x97df, seq=3/768, ttl=61 (request in 606)
608 4552.780958 10.0.10.10 10.0.11.10 ICMP 98 Echo (ping) request id=0x98df, seq=4/1024, ttl=64 (reply in 609)
609 4552.796619 10.0.11.10 10.0.10.10 ICMP 98 Echo (ping) reply id=0x98df, seq=4/1024, ttl=61 (request in 608)

Frame 606: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
  Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)
    Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)
      .... ..0. .... = LG bit: Globally unique address (factory default)
      .... ...0 .... = IG bit: Individual address (unicast)
    Source: Private_66:68:00 (00:50:79:66:68:00)
      .... ..0. .... = LG bit: Globally unique address (factory default)
      .... ...0 .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 8]
  Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0xdf97 (57239)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0x71fe [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.10.10
```

Рис. 70: Захват трафика OSPF (пакет 3)

615	4558.457043	10.0.10.10	10.0.11.10	TCP	74	5616 → 5617 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765793694 TSecr=0 WS=2		
616	4558.458884	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)		
617	4558.460457	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 5616 → 5617 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=176579369		
▶	Frame 615: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0							0000 0c 94 80 bd 00 00 00
▼	Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0010 00 3c 00 00 40 00 01
▼	Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0020 0b 0a 15 f0 15 f1 69
	 0. = LG bit: Globally unique address (factory default)							0030 00 00 64 56 00 00 02
	 0 = IG bit: Individual address (unicast)							0040 df 9e 00 00 00 00 01
▼	Source: Private_66:68:00 (00:50:79:66:68:00)							
	 0. = LG bit: Globally unique address (factory default)							
	 0 = IG bit: Individual address (unicast)							
	Type: IPv4 (0x0800)							
	[Stream index: 8]							
▼	Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10							
	0100 = Version: 4							
	 0101 = Header Length: 20 bytes (5)							
▶	Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)							
	Total Length: 60							
	Identification: 0x0000 (0)							
▶	010. = Flags: 0x2, Don't fragment							
	...0 0000 0000 0000 = Fragment Offset: 0							
▼	Time to Live: 1							
▶	[Expert Info (Note/Sequence): "Time To Live" only 1]							
	Protocol: TCP (6)							
	Header Checksum: 0x50a9 [validation disabled]							
	[Header checksum status: Unverified]							
	Source Address: 10.0.10.10							
	Destination Address: 10.0.11.10							
	[Stream index: 2]							
▶	Transmission Control Protocol, Src Port: 5616, Dst Port: 5617, Seq: 0, Len: 0							

Рис. 71: Захват трафика OSPF (пакет 4)


```
→ 361 4792.725291 10.0.10.10 10.0.11.10 ICMP 98 Echo (ping) request id=0x3cdf, seq=4/1024, ttl=61 (reply in 362)
← 362 4792.727716 10.0.11.10 10.0.10.10 ICMP 98 Echo (ping) reply id=0x3cdf, seq=4/1024, ttl=64 (request in 361)

▶ Frame 361: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
▼ Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
  ▼ Destination: Private_66:68:01 (00:50:79:66:68:01)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 9]
  ▼ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 84
      Identification: 0xdf3c (57148)
    ▶ 000. .... = Flags: 0x0
      ...0 0000 0000 0000 = Fragment Offset: 0
      Time to Live: 61
      Protocol: ICMP (1)
      Header Checksum: 0x7559 [validation disabled]
      [Header checksum status: Unverified]
      Source Address: 10.0.10.10
      Destination Address: 10.0.11.10
      [Stream index: 2]
  ▶ Internet Control Message Protocol
```

0000	00 50 79 66 68 01
0010	00 54 df 3c 00 00
0020	0b 0a 08 00 e3 28
0030	0e 0f 10 11 12 13
0040	1e 1f 20 21 22 23
0050	2e 2f 30 31 32 33
0060	3e 3f

Рис. 72: Захват трафика OSPF (пакет 5)

```

367 4796.802934 10.0.10.10 10.0.11.10 TCP 74 19073 → 19074 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765793600 TSecr=0 WS=2
368 4796.803356 10.0.11.10 10.0.10.10 TCP 54 19074 → 19073 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
369 4796.810304 10.0.10.10 10.0.11.10 TCP 74 [TCP Retransmission] 19073 → 19074 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765

```

▸ Frame 367: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0

▾ Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)

- ▾ Destination: Private_66:68:01 (00:50:79:66:68:01)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)
- ▾ Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
 -0. = LG bit: Globally unique address (factory default)
 -0. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)
[Stream index: 9]

▾ Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10

0100 = Version: 4
 ... 0101 = Header Length: 20 bytes (5)

▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 60
 Identification: 0x0009 (9)

▸ 010. = Flags: 0x2, Don't fragment
 ...0 0000 0000 0000 = Fragment Offset: 0

▸ Time to Live: 1
 Protocol: TCP (6)
 Header Checksum: 0x50a0 [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 10.0.10.10
 Destination Address: 10.0.11.10
 [Stream index: 2]

▸ Transmission Control Protocol, Src Port: 19073, Dst Port: 19074, Seq: 0, Len: 0

Рис. 73: Захват трафика OSPF (пакет 6)

→	393	4883.265669	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x97df, seq=3/768, ttl=61 (reply in 394)		
←	394	4883.266693	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x97df, seq=3/768, ttl=64 (request in 393)		
	395	4884.267015	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x97df, seq=3/768, ttl=61 (reply in 396)		

▶	Frame 393: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0							0000	00 50 79 66 68 01
▼	Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)							0010	00 54 df 97 00 00
▼	Destination: Private_66:68:01 (00:50:79:66:68:01)							0020	0b 0a 08 00 88 29
....	..0. = LG bit: Globally unique address (factory default)							0030	0e 0f 10 11 12 13
....	...0 = IG bit: Individual address (unicast)							0040	1e 1f 20 21 22 23
▼	Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)							0050	2e 2f 30 31 32 33
....	..0. = LG bit: Globally unique address (factory default)							0060	3e 3f
....	...0 = IG bit: Individual address (unicast)								
	Type: IPv4 (0x0800)								
	[Stream index: 9]								
▼	Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10								
	0100 = Version: 4								
	 0101 = Header Length: 20 bytes (5)								
▶	Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)								
	Total Length: 84								
	Identification: 0xdf97 (57239)								
▶	000. = Flags: 0x0								
	...0 0000 0000 0000 = Fragment Offset: 0								
	Time to Live: 61								
	Protocol: ICMP (1)								
	Header Checksum: 0x74fe [validation disabled]								
	[Header checksum status: Unverified]								
	Source Address: 10.0.10.10								
	Destination Address: 10.0.11.10								
	[Stream index: 2]								
▶	Internet Control Message Protocol								

Рис. 74: Захват трафика OSPF (пакет 7)

403 4890.093478 10.0.10.10 10.0.11.10 TCP 74 5616 → 5617 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765793694 TSecr=0 WS=2
404 4890.094730 10.0.11.10 10.0.10.10 TCP 54 5617 → 5616 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
405 4890.103360 10.0.10.10 10.0.11.10 TCP 74 [TCP Retransmission] 5616 → 5617 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1765793694

Frame 403: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface -, id 0

Ethernet II, Src: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00), Dst: Private_66:68:01 (00:50:79:66:68:01)

- Destination: Private_66:68:01 (00:50:79:66:68:01)
 - 0. = LG bit: Globally unique address (factory default)
 - 0 = IG bit: Individual address (unicast)
- Source: 0c:f5:7e:d9:00:00 (0c:f5:7e:d9:00:00)
 - 0. = LG bit: Globally unique address (factory default)
 - 0 = IG bit: Individual address (unicast)
- Type: IPv4 (0x0800)
- [Stream index: 9]

Internet Protocol Version 4, Src: 10.0.10.10, Dst: 10.0.11.10

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 60
- Identification: 0x0009 (9)
- 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 1
 - Protocol: TCP (6)
 - Header Checksum: 0x50a0 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 10.0.10.10
 - Destination Address: 10.0.11.10
 - [Stream index: 2]

Transmission Control Protocol, Src Port: 5616, Dst Port: 5617, Seq: 0, Len: 0

- Source Port: 5616
- Destination Port: 5617
- [Stream index: 9]
- [Stream Packet Number: 1]
- [Conversation completeness: Incomplete, CLIENT_ESTABLISHED (3)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 1765793694

0000 00 50 79 66 68 01 0c f5 7e
0010 00 3c 00 09 40 00 01 06 50
0020 0b 0a 15 f0 15 f1 69 3f df
0030 00 00 64 56 00 00 02 04 05
0040 df 9e 00 00 00 00 01 03 03

Рис. 75: Захват трафика OSPF (пакет 8)

Успешный ICMP-обмен (запрос-ответ) с TTL=61/64.

Пакеты с TTL=1 не проходят дальше первого маршрутизатора.

TCP-сессии устанавливаются успешно (SYN, SYN-ACK).

Видна инкапсуляция и маршрутизация на уровне L3.

10. На маршрутизаторах настройте OSPFv3 для сетей IPv6:

```
msk-dsadova-gw-01# configure terminal
msk-dsadova-gw-01(config)# router ospf6
msk-dsadova-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-dsadova-gw-01(config-ospf6)# exit
msk-dsadova-gw-01(config)# interface eth0
msk-dsadova-gw-01(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth1
msk-dsadova-gw-01(config-if)#  ipv6 ospf6 area 0
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)#  interface eth2
msk-dsadova-gw-01(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# exit
msk-dsadova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-01#
```

```
msk-dsadova-gw-02# configure terminal
msk-dsadova-gw-02(config)# router ospf6
msk-dsadova-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-dsadova-gw-02(config-ospf6)# exit
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# interface eth1
msk-dsadova-gw-02(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 77: Настройка OSPFv3

```
msk-dsadova-gw-03# configure terminal
msk-dsadova-gw-03(config)# router ospf6
msk-dsadova-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-dsadova-gw-03(config-ospf6)# exit
msk-dsadova-gw-03(config)# interface eth0
msk-dsadova-gw-03(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# interface eth1
msk-dsadova-gw-03(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# interface eth2
msk-dsadova-gw-03(config-if)# ipv6 ospf6 area 0
msk-dsadova-gw-03(config-if)# exit
msk-dsadova-gw-03(config)# exit
msk-dsadova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-03#
```

Рис. 78: Настройка OSPFv3


```
msk-dsadova-gw-04# configure terminal
msk-dsadova-gw-04(config)# router ospf6
msk-dsadova-gw-04(config-ospf6)#  ospf6 router-id 4.4.4.4
msk-dsadova-gw-04(config-ospf6)# exit
msk-dsadova-gw-04(config)# interface eth0
msk-dsadova-gw-04(config-if)#  ipv6 ospf6 area 0
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# interface eth1
msk-dsadova-gw-04(config-if)#  ipv6 ospf6 area 0
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# exit
msk-dsadova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-04#
```

Рис. 79: Настройка OSPFv3

11. С PC1 пропингуйте PC2 и определите путь следования пакетов:

```
PC1-dsadova> ping 2001:11::a
```

```
2001:11::a icmp6_seq=1 ttl=58 time=23.614 ms
```

```
2001:11::a icmp6_seq=2 ttl=58 time=24.412 ms
```

```
2001:11::a icmp6_seq=3 ttl=58 time=13.367 ms
```

```
2001:11::a icmp6_seq=4 ttl=58 time=38.317 ms
```

```
2001:11::a icmp6_seq=5 ttl=58 time=18.548 ms
```

```
PC1-dsadova> trace 2001:11::a
```

```
trace to 2001:11::a, 64 hops max
```

```
 1 2001:10::1      5.431 ms   1.075 ms   1.110 ms
```

```
 2 2001:4::1       14.576 ms  8.962 ms   4.552 ms
```

```
 3 2001:3::1       15.421 ms  21.170 ms   8.302 ms
```

```
 4 2001:11::a      10.082 ms  8.186 ms   12.472 ms
```

```
PC1-dsadova> 
```

12. Проверьте таблицу маршрутизации протокола OSPFv3:

```
msk-dsadova-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri      DeadTime      State/IfState      Duration I/F[State]
2.2.2.2          1        00:00:30      Full/BDR           00:04:50 eth1[DR]
4.4.4.4          1        00:00:36      Full/BDR           00:00:50 eth2[DR]
msk-dsadova-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::                                eth1 00:00:52
*N IA 2001:2::/64      fe80::e74:9dff:febf:0           eth1 00:00:52
*N IA 2001:3::/64      fe80::e0c:4aff:febf:1           eth2 00:00:52
*N IA 2001:4::/64      ::                                eth2 00:00:52
*N IA 2001:10::/64     ::                                eth0 00:00:52
*N IA 2001:11::/64     fe80::e74:9dff:febf:0           eth1 00:00:52
                        fe80::e0c:4aff:febf:1           eth2
msk-dsadova-gw-01#
```

Рис. 81: Таблица OSPFv3 на gw-01

13. Предположим, что пакет проходит через маршрутизатор msk-user-gw-02 (в противном случае для дальнейших действий используйте маршрутизатор msk-user-gw-04). Отключите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02#  
msk-dsadova-gw-02#  configure terminal  
msk-dsadova-gw-02(config)#  interface eth0  
msk-dsadova-gw-02(config-if)#  shutdown  
msk-dsadova-gw-02(config-if)#
```

Рис. 82: Отключение интерфейса на gw-02 (OSPFv3)

14. Проверьте таблицу маршрутизации протокола OSPFv3:

```
mzk-dsadova-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64          ::          eth1 00:00:17
*N IA 2001:2::/64          fe80::e0c:4aff:fef8:1  eth2 00:00:17
*N IA 2001:3::/64          fe80::e0c:4aff:fef8:1  eth2 00:00:17
*N IA 2001:4::/64          ::          eth2 00:00:17
*N IA 2001:10::/64         ::          eth0 00:00:17
*N IA 2001:11::/64         fe80::e0c:4aff:fef8:1  eth2 00:00:17
mzk-dsadova-gw-01#
```

Рис. 83: Таблица OSPFv3 после отключения интерфейса

15. С PC1 пропируйте PC2 и определите путь следования пакетов. Отметьте примерное время, за которое восстановится маршрут.

```
PC1-dsadova> trace 2001:11::a
```

```
trace to 2001:11::a, 64 hops max
```

```
 1 2001:10::1    18.953 ms  5.652 ms  5.030 ms
 2 2001:4::1     6.098 ms   7.635 ms  17.124 ms
 3 2001:3::1     8.093 ms   9.047 ms  11.260 ms
 4 2001:11::a   48.115 ms  17.612 ms  13.606 ms
```

```
PC1-dsadova> ping 2001:11::a
```

```
2001:11::a icmp6_seq=1 ttl=58 time=26.184 ms
2001:11::a icmp6_seq=2 ttl=58 time=21.515 ms
2001:11::a icmp6_seq=3 ttl=58 time=13.711 ms
2001:11::a icmp6_seq=4 ttl=58 time=13.964 ms
2001:11::a icmp6_seq=5 ttl=58 time=13.050 ms
```

Traceroute показывает путь через маршрутизаторы: 2001:10::1, 2001:4::1, 2001:3::1, 2001:11::a

Время отклика: 10-30 мс.

16. Включите на маршрутизаторе msk-user-gw-02 интерфейс:

```
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# exit
msk-dsadova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-dsadova-gw-02#
```

Рис. 85: Включение интерфейса на gw-02 (OSPFv3)

17. С PC1 пропингуйте PC2 и определите путь следования пакетов.

```
PC1-dsadova> ping 2001:11::a
```

```
2001:11::a icmp6_seq=1 ttl=58 time=32.693 ms
```

```
2001:11::a icmp6_seq=2 ttl=58 time=15.881 ms
```

```
2001:11::a icmp6_seq=3 ttl=58 time=15.355 ms
```

```
2001:11::a icmp6_seq=4 ttl=58 time=18.617 ms
```

```
2001:11::a icmp6_seq=5 ttl=58 time=10.321 ms
```

```
PC1-dsadova> trace 2001:11::a
```

```
trace to 2001:11::a, 64 hops max
```

```
 1 2001:10::1    3.709 ms   6.583 ms   3.226 ms
```

```
 2 2001:4::1     5.307 ms  14.923 ms   7.976 ms
```

```
 3 2001:3::1     8.860 ms  12.033 ms   9.938 ms
```

```
 4 2001:11::a    5.640 ms  24.311 ms   7.009 ms
```

```
PC1-dsadova> 
```

18. Посмотрите захваченный на соединениях трафик. В отчёте поясните, какую информацию можно извлечь из перехваченных пакетов.

845	5282.604174	2001:10::a	2001:11::a	UDP	126 62513 → 62514 Len=64		
846	5282.618073	2001:10::1	ff02::1:ff00:a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:94:80:bd:00:00		
847	5282.619064	2001:10::a	2001:10::1	ICMPv6	86 Neighbor Advertisement 2001:10::a (sol, ovr) is at 00:50:79:66:68:00		
▼ Frame 845: Packet, 126 bytes on wire (1008 bits), 126 bytes captured (1008 bits) on interface -, id 0							0000 0c 94 80 1
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0010 00 00 00 1
▼ Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0020 00 00 00 0
.....0. = LG bit: Globally unique address (factory default)							0030 00 00 00 0
.....0. = IG bit: Individual address (unicast)							0040 79 66 68 0
▼ Source: Private_66:68:00 (00:50:79:66:68:00)							0050 1a 1b 1c 1
.....0. = LG bit: Globally unique address (factory default)							0060 2a 2b 2c 1
.....0. = IG bit: Individual address (unicast)							0070 3a 3b 3c 1
Type: IPv6 (0x86dd)							
[Stream index: 8]							
▼ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a							
▼ User Datagram Protocol, Src Port: 62513, Dst Port: 62514							
▼ Data (64 bytes)							

Рис. 87: Захват трафика OSPFv3

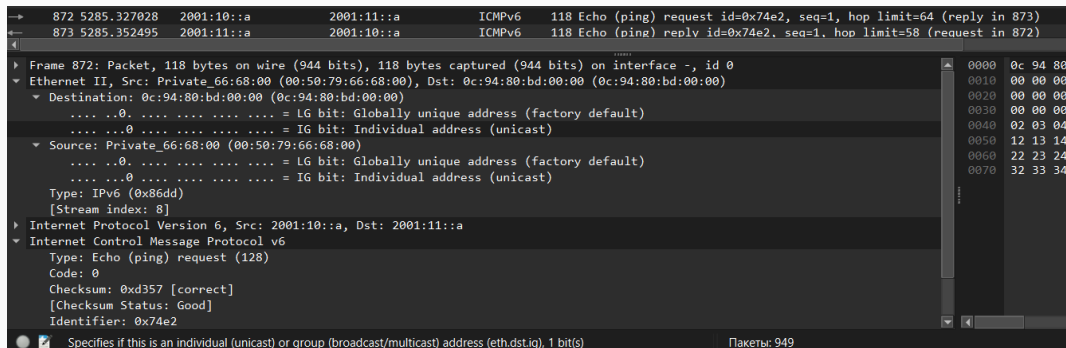


Рис. 88: Захват трафика OSPFv3

→	872	5285.327028	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x74e2, seq=1, hop limit=64 (reply in 873)	
←	873	5285.352495	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x74e2, seq=1, hop limit=58 (request in 872)	
- Frame 872: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0 - Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00) - Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00) 0. = LG bit: Globally unique address (factory default) 0. = IG bit: Individual address (unicast) - Source: Private_66:68:00 (00:50:79:66:68:00) 0. = LG bit: Globally unique address (factory default) 0. = IG bit: Individual address (unicast) - Type: IPv6 (0x86dd) [Stream index: 8] - Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a - Internet Control Message Protocol v6 - Type: Echo (ping) request (128) - Code: 0 - Checksum: 0xd357 [correct] [Checksum Status: Good] - Identifier: 0x74e2 - Sequence: 1 <u>[Response In: 873]</u> - Data (56 bytes)							0000 0c 94 80 0010 00 00 00 0020 00 00 00 0030 00 00 00 0040 02 03 04 0050 12 13 14 0060 22 23 24 0070 32 33 34

Рис. 89: Захват трафика OSPFv3

→	898	5348.360124	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0xb3e2, seq=1, hop limit=64 (reply in 899)		
←	899	5348.392225	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0xb3e2, seq=1, hop limit=58 (request in 898)		
▼								
▶ Frame 898: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0							0000	0c 94
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0010	00 00
▼ Destination: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)							0020	00 00
.... ..0. = LG bit: Globally unique address (factory default)							0030	00 00
.... ...0 = IG bit: Individual address (unicast)							0040	02 03
▼ Source: Private_66:68:00 (00:50:79:66:68:00)							0050	12 13
.... ..0. = LG bit: Globally unique address (factory default)							0060	22 23
.... ...0 = IG bit: Individual address (unicast)							0070	32 33
Type: IPv6 (0x86dd)								
[Stream index: 8]								
▶ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a								
▼ Internet Control Message Protocol v6								
Type: Echo (ping) request (128)								
Code: 0								
Checksum: 0x9457 [correct]								
[Checksum Status: Good]								
Identifier: 0xb3e2								
Sequence: 1								
[Response In: 899]								
▶ Data (56 bytes)								

Рис. 90: Захват трафика OSPFv3

913	5355.423634	2001:10::a	2001:11::a	UDP	126 22960 → 22961 Len=64	
914	5355.425976	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)	
915	5355.427319	2001:10::a	2001:11::a	UDP	126 22960 → 22961 Len=64	
▶ Frame 914: Packet, 174 bytes on wire (1392 bits), 174 bytes captured (1392 bits) on interface -, id 0						0000
▼ Ethernet II, Src: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)						0010
▼ Destination: Private_66:68:00 (00:50:79:66:68:00)						0020
.... ..0. = LG bit: Globally unique address (factory default)						0030
....0 = IG bit: Individual address (unicast)						0040
▼ Source: 0c:94:80:bd:00:00 (0c:94:80:bd:00:00)						0050
.... ..0. = LG bit: Globally unique address (factory default)						0060
....0 = IG bit: Individual address (unicast)						0070
Type: IPv6 (0x86dd)						0080
[Stream index: 8]						0090
▶ Internet Protocol Version 6, Src: 2001:10::1, Dst: 2001:10::a						00a0
▼ Internet Control Message Protocol v6						
▼ Type: Time Exceeded (3)						
▶ [Expert Info (Note/Response): Type indicates an error]						
Code: 0 (Hop limit exceeded in transit)						
Checksum: 0x4b30 [correct]						
[Checksum Status: Good]						
Reserved: 00000000						
▶ Internet Protocol Version 6, Src: 2001:10::a, Dst: 2001:11::a						
▶ User Datagram Protocol, Src Port: 22960, Dst Port: 22961						
▶ Data (64 bytes)						

Рис. 91: Захват трафика OSPFv3

Успешный обмен ICMPv6.

TCP-сессии устанавливаются.

TTL/HL корректно уменьшаются.

Видна работа двойного стека (IPv4 и IPv6).

Построение туннеля IPv6-IPv4

Постановка задачи

Заданы две IPv6-сети и IPv4-сеть. Требуется организовать туннель IPv6 поверх IPv4, позволяющий передавать данные из одной IPv6-сети в другую IPv6-сеть через сеть IPv4. Назначаемые устройствам сети адреса указаны в табл. 8.3

При организации туннеля IPv6 поверх IPv4 трафик IPv6 отправляется внутри пакетов IPv4, в заголовках которых номер протокола IP имеет значение 41. Накладные расходы инкапсуляции состоят в том, что размер заголовка IPv4 имеет размер 20 байт, соответственно при MTU в 1500 байт IPv6-пакеты размером 1480 байт можно отправлять без фрагментации.

1. Создайте новый проект в GNS3.
2. В рабочем пространстве разместите и соедините устройства в соответствии с топологией, приведённой на рис. 8.5. Используйте маршрутизаторы VyOS.

3. Измените отображаемые названия устройств. Коммутаторам присвойте названия по принципу `msk-user-sw-0x`, маршрутизаторам — по принципу `msk-user-gw-0x`, VPCS — по принципу `PCx-user`, где вместо `user` укажите имя вашей учётной записи, вместо `x` — порядковый номер устройства.

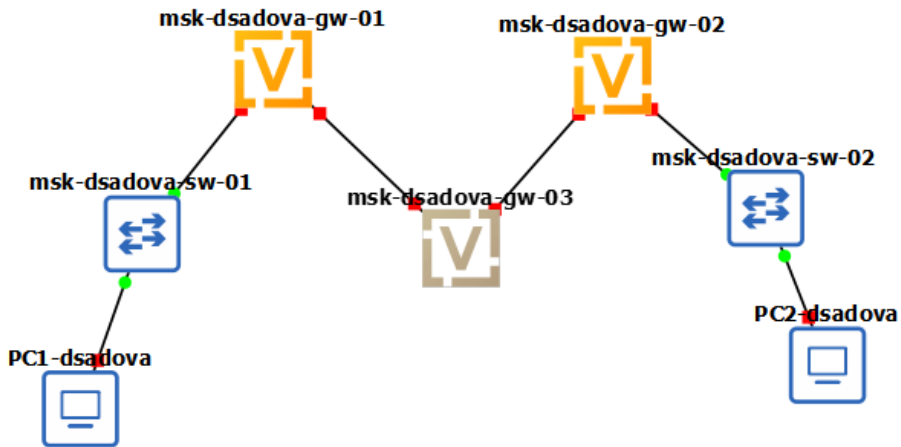


Рис. 92: Топология туннеля IPv6-IPv4

4. На соединении между первым и третьим маршрутизаторами подключите анализатор трафика.

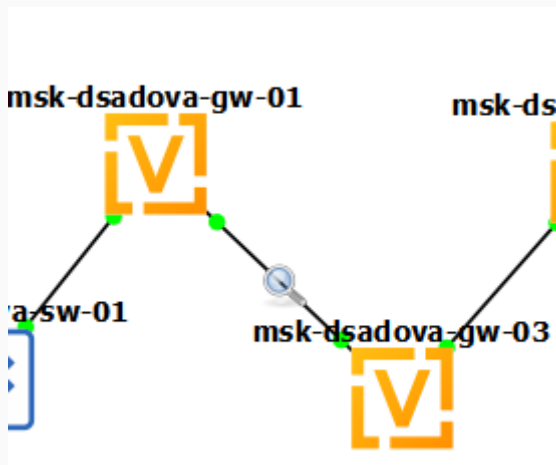


Рис. 93: Подключение анализатора трафика

5. Присвойте адреса оконечным устройствам PC1 и PC2 в соответствии с данными в табл. 8.3:

```
PC1-dsadova> ip 1000::a/64
PC1 : 1000::a/64

PC1-dsadova> save
Saving startup configuration to startup.vpc
. done

PC1-dsadova> show ipv6

NAME                : PC1-dsadova[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:00
LPORT                : 10008
RHOST:PORT           : 127.0.0.1:10009
MTU:                 : 1500
```

```
PC2-dsadova> ip 1002::a/64
PC1 : 1002::a/64

PC2-dsadova> save
Saving startup configuration to startup.vpc
. done

PC2-dsadova> show ipv6

NAME                : PC2-dsadova[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:01
LPORT                : 10010
RHOST:PORT           : 127.0.0.1:10011
MTU                  : 1500

PC2-dsadova> 
```

Рис. 95: Настройка адресов на PC2 (туннель)

6. Установите систему на маршрутизаторы VyOS.
7. На маршрутизаторах перейдите в режим конфигурирования, измените имя устройства (вместо user укажите имя вашей учётной записи, вместо x - порядковый номер устройства):


```
vyos@vyos:~$ configure
set system host-name msk-user-gw-0x[edit]
vyos@vyos# set system host-name msk-dsadova-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dsadova-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] ☐
```

Рис. 96: Настройка имени маршрутизатора R1

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dsadova-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dsadova-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-dsadova-gw-03
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-dsadova-gw-03
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```

Рис. 98: Настройка имени маршрутизатора R3

8. Настройте адреса на интерфейсах маршрутизаторов:

```
vyos@msk-dsadova-gw-01:~$ configure
[edit]
vyos@msk-dsadova-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-dsadova-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
vyos@msk-dsadova-gw-01#
[edit]
vyos@msk-dsadova-gw-01# set service router-advert interface eth0 prefix 1000::/64
4
[edit]
vyos@msk-dsadova-gw-01# commit
[edit]
vyos@msk-dsadova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-01#
```

Рис. 99: Настройка адресов на R1

```
vyos@msk-dsadova-gw-02:~$ configure

[edit]
vyos@msk-dsadova-gw-02#
[edit]
vyos@msk-dsadova-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-dsadova-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-dsadova-gw-02# set service router-advert interface eth0 prefix 1002::/64
4
[edit]
vyos@msk-dsadova-gw-02# commit
[edit]
vyos@msk-dsadova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
```

Рис. 100: Настройка адресов на R2

```
vyos@msk-dsadova-gw-03# show interfaces ethernet eth0
  address dhcp
+address 10.0.0.2/8
  hw-id 0c:6d:5d:48:00:00
[edit]
vyos@msk-dsadova-gw-03# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-dsadova-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8

  Configuration path: [interfaces ethernet eth0 address 10.0.0.2/8] already exists
[edit]
vyos@msk-dsadova-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8

  Configuration path: [interfaces ethernet eth1 address 20.0.0.1/8] already exists
[edit]
vyos@msk-dsadova-gw-03# commit
[edit]
vyos@msk-dsadova-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-03#
```

Рис. 101: Настройка адресов на R3

9. Убедитесь, что на PC появились адреса ближайших к ним маршрутизаторов:

```
PC1-dsadova> show ipv6

NAME                : PC1-dsadova[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                  :
ROUTER LINK-LAYER   : 0c:15:cd:16:00:00
MAC                  : 00:50:79:66:68:00
LPORT                : 10012
RHOST:PORT           : 127.0.0.1:10013
MTU                  : 1500

PC1-dsadova> 
```

10. Проверьте маршруты с маршрутизатора R1:

```
vyos@msk-dsadova-gw-01# ping 20.0.0.1
connect: Network is unreachable
[edit]
vyos@msk-dsadova-gw-01# ping 20.0.0.2
connect: Network is unreachable
[edit]
vyos@msk-dsadova-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.42 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.48 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.59 ms
```

Рис. 103: Таблица маршрутизации на R1

В отчёте поясните результат проверки, в том числе с учётом анализа трафика.

→	183	86.242758	10.0.0.1	10.0.0.2	ICMP	98 Echo (ping) request	id=0x10a2, seq=87/22272, ttl=64 (reply in 184)		
←	184	86.245259	10.0.0.2	10.0.0.1	ICMP	98 Echo (ping) reply	id=0x10a2, seq=87/22272, ttl=64 (request in 183)		
▶ Frame 183: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0									
▶ Ethernet II, Src: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01), Dst: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00)									
▶ Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.2									
▶ Internet Control Message Protocol									
								0000	0c 6d 5d
								0010	00 54 6c
								0020	00 02 08
								0030	00 00 a4

Рис. 104: Анализ трафика до настройки туннеля

11. Настройте маршрутизацию IPv4:

```
vyos@msk-dsadova-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-dsadova-gw-01# commit
[edit]
vyos@msk-dsadova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-01#
```

Рис. 105: Настройка статической маршрутизации IPv4 на R1

```
vyos@msk-dsadova-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-dsadova-gw-02# commit
[edit]
vyos@msk-dsadova-gw-02#
[edit]
vyos@msk-dsadova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-02#
```

Рис. 106: Настройка статической маршрутизации IPv4 на R2

```
vyos@msk-dsadova-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-dsadova-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-dsadova-gw-03# commit
[edit]
vyos@msk-dsadova-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-03#
```

Рис. 107: Настройка статической маршрутизации IPv4 на R3

12. Проверьте маршруты:

```
vyos@misk-dsadova-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=6.92 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.69 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=3.32 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=2.59 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=3.03 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=2.05 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=1.88 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=2.32 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=3.17 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=2.83 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=2.64 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=2.20 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=1.79 ms
^C
--- 10.0.0.2 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 45ms
rtt min/avg/max/mdev = 1.690/2.803/6.918/1.293 ms
[edit]
vyos@misk-dsadova-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=3.14 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=2.00 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=2.99 ms
64 bytes from 20.0.0.1: icmp_seq=4 ttl=64 time=2.40 ms
64 bytes from 20.0.0.1: icmp_seq=5 ttl=64 time=2.28 ms
64 bytes from 20.0.0.1: icmp_seq=6 ttl=64 time=1.94 ms
64 bytes from 20.0.0.1: icmp_seq=7 ttl=64 time=2.57 ms
64 bytes from 20.0.0.1: icmp_seq=8 ttl=64 time=3.15 ms
64 bytes from 20.0.0.1: icmp_seq=9 ttl=64 time=4.86 ms
^X64 bytes from 20.0.0.1: icmp_seq=10 ttl=64 time=1.76 ms
^C
--- 20.0.0.1 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 36ms
rtt min/avg/max/mdev = 1.763/2.708/4.860/0.859 ms
[edit]
vyos@misk-dsadova-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=5.53 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=3.57 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=3.69 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=3.67 ms
^C
--- 20.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 10ms
rtt min/avg/max/mdev = 3.569/4.115/5.531/0.821 ms
[edit]
vyos@misk-dsadova-gw-01#
```

В отчёте поясните результат проверки, в том числе с учётом анализа трафика.

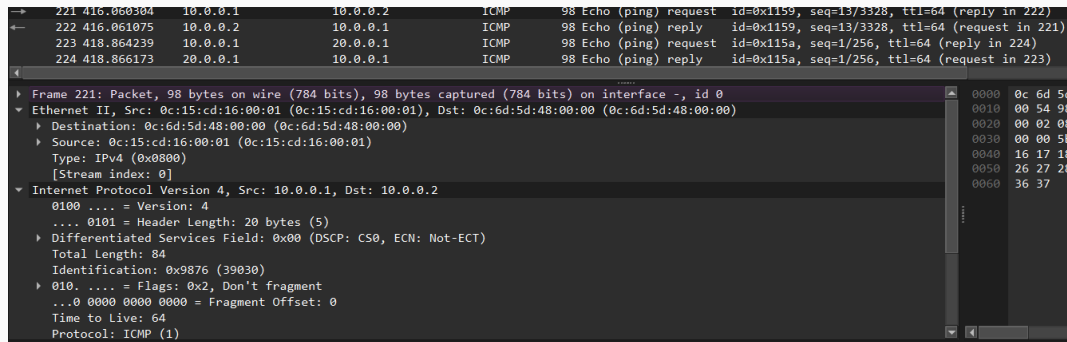


Рис. 109: Анализ трафика IPv4 (пакет 1)

→	225	419.868224	10.0.0.1	20.0.0.1	ICMP	98 Echo (ping) request	id=0x115a, seq=2/512, ttl=64 (reply in 226)
←	226	419.869375	20.0.0.1	10.0.0.1	ICMP	98 Echo (ping) reply	id=0x115a, seq=2/512, ttl=64 (request in 225)
▶ Frame 225: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0 ▼ Ethernet II, Src: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01), Dst: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00) ▶ Destination: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00) ▶ Source: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01) Type: IPv4 (0x0800) [Stream index: 0] ▼ Internet Protocol Version 4, Src: 10.0.0.1, Dst: 20.0.0.1 0100 = Version: 4 0101 = Header Length: 20 bytes (5) ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) Total Length: 84 Identification: 0xc99c (51612) ▶ 010. = Flags: 0x2, Don't fragment ...0 0000 0000 0000 = Fragment Offset: 0 Time to Live: 64 Protocol: ICMP (1) Header Checksum: 0x530b [validation disabled] 0000 0c 6d 5d 0010 00 54 c9 0020 00 01 08 0030 00 00 6e 0040 16 17 18 0050 26 27 28 0060 36 37							

Рис. 110: Анализ трафика IPv4 (пакет 2)

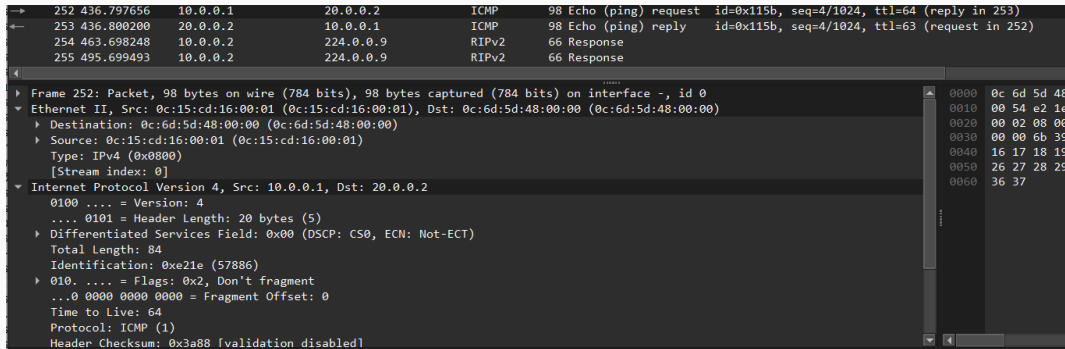


Рис. 111: Анализ трафика IPv4 (пакет 3)

Работает только IPv4 маршрутизация. Ping между 10.0.0.1 и 10.0.0.2 успешен

RIPv2-пакеты отправляются на multicast-адрес 224.0.0.9

Периодичность обновлений: примерно каждые 30 секунд

13. Создайте туннель IPv6 через сеть IPv4:

```
vyos@msk-dsadova-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dsadova-gw-01# set interfaces tunnel tun0 source-address 10.0.0.1
[edit]
vyos@msk-dsadova-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-dsadova-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-dsadova-gw-01# commit
[edit]
vyos@msk-dsadova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-01#
```

Рис. 112: Настройка туннеля на R1

```
vyos@msk-dsadova-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-dsadova-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-dsadova-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-dsadova-gw-02# set interfaces tunnel tun0 address 1001::2/64

[edit]
vyos@msk-dsadova-gw-02#
[edit]
vyos@msk-dsadova-gw-02# commit
[edit]
vyos@msk-dsadova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-02#
```

Рис. 113: Настройка туннеля на R3

14. Настройте статическую маршрутизацию IPv6:

```
vyos@msk-dsadova-gw-01# set protocols static route6 1002::0/64 next-hop 1001::2
[edit]
vyos@msk-dsadova-gw-01# commit
[edit]
vyos@msk-dsadova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-01#
```

Рис. 114: Настройка маршрутизации IPv6 на R1

```
vyos@msk-dsadova-gw-02# set protocols static route6 1000::0/64 next-hop 1001::1
[edit]
vyos@msk-dsadova-gw-02# commit
save
[edit]
vyos@msk-dsadova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-dsadova-gw-02#
```

Рис. 115: Настройка маршрутизации IPv6 на R3

15. Проверьте доступность оконечных устройств:

```
PC1-dsadova> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=48.511 ms
v1002::a icmp6_seq=2 ttl=60 time=5.489 ms
002::a icmp6_seq=3 ttl=60 time=7.348 ms
1002::a icmp6_seq=4 ttl=60 time=5.784 ms
1002::a icmp6_seq=5 ttl=60 time=7.276 ms

PC1-dsadova> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1      8.802 ms   1.701 ms   1.238 ms
 2 1001::2      8.958 ms   4.169 ms   3.182 ms
 3 1002::a      4.303 ms   5.164 ms   2.852 ms

PC1-dsadova> 
```

```
PC2-dsadova> ping 1000::a
```

```
1000::a icmp6_seq=1 ttl=60 time=9.826 ms
```

```
1000::a icmp6_seq=2 ttl=60 time=7.559 ms
```

```
1000::a icmp6_seq=3 ttl=60 time=8.757 ms
```

```
1000::a icmp6_seq=4 ttl=60 time=6.902 ms
```

```
1000::a icmp6_seq=5 ttl=60 time=6.305 ms
```

```
PC2-dsadova> trace 1000::a
```

```
trace to 1000::a, 64 hops max
```

```
1 1002::1      1.669 ms    1.549 ms    1.244 ms
```

```
2 1001::1      5.064 ms    2.672 ms    7.403 ms
```

```
3 1000::a      7.369 ms    6.820 ms    3.930 ms
```

```
PC2-dsadova> █
```

Рис. 117: Ping с PC2 на PC1 через туннель

В отчёте поясните результат проверки. Используя данные, собранные анализатором трафика, поясните, где и в каком виде содержится информация о прохождении пакетов по туннелю.

→	275	859.015752	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0x2a60, seq=5, hop limit=63 (reply in 276)		
←	276	859.019890	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0x2a60, seq=5, hop limit=61 (request in 275)		
	277	859.986743	0c:6d:5d:48:00:00	0c:15:cd:16:00:01	ARP	60 Who has 10.0.0.1? Tell 10.0.0.2		
	278	859.988824	0c:15:cd:16:00:01	0c:6d:5d:48:00:00	ARP	60 10.0.0.1 is at 0c:15:cd:16:00:01		
	279	860.175553	0c:15:cd:16:00:01	0c:6d:5d:48:00:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1		
	280	860.176735	0c:6d:5d:48:00:00	0c:15:cd:16:00:01	ARP	60 10.0.0.2 is at 0c:6d:5d:48:00:00		
▶	Frame 275: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0						0000	0c 6d
▼	Ethernet II, Src: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01), Dst: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00)						0010	00 7c
▶	Destination: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00)						0020	00 02
▶	Source: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01)						0030	00 00
	Type: IPv4 (0x0800)						0040	00 00
	[Stream index: 0]						0050	00 05
▼	Internet Protocol Version 4, Src: 10.0.0.1, Dst: 20.0.0.2						0060	0e 0f
	0100 = Version: 4						0070	1e 1f
	 0101 = Header Length: 20 bytes (5)						0080	2e 2f
▶	Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)							
	Total Length: 124							
	Identification: 0xd987 (55687)							
▶	010. = Flags: 0x2, Don't fragment							
	...0 0000 0000 0000 = Fragment Offset: 0							
	Time to Live: 64							
	Protocol: IPv6 (41)							
	Header Checksum: 0x42cf [validation disabled]							
	[Header checksum status: Unverified]							
	Source Address: 10.0.0.1							
	Destination Address: 20.0.0.2							
	[Stream index: 6]							
▶	Internet Protocol Version 6, Src: 1000::a, Dst: 1002::a							
▶	Internet Control Message Protocol v6							

300	879.197325	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0x3f60, seq=4, hop limit=63 (reply in 301)				
301	879.199907	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0x3f60, seq=4, hop limit=61 (request in 300)				
302	880.204420	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0x3f60, seq=5, hop limit=63 (reply in 303)				
303	880.206960	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0x3f60, seq=5, hop limit=61 (request in 302)				
304	881.838384	1002::a	1000::a	UDP	146 63488 → 63489 Len=64				
305	881.839830	1001::1	1002::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)				
306	881.842632	1002::a	1000::a	UDP	146 63488 → 63489 Len=64				
307	881.843046	1001::1	1002::a	ICMPv6	194 Time Exceeded (Hop limit exceeded in transit)				
308	881.847097	1002::a	1000::a	UDP	146 63488 → 63489 Len=64				
Frame 301: Packet, 138 bytes on wire (1104 bits), 138 bytes captured (1104 bits) on interface -, id 0									
Ethernet II, Src: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01), Dst: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00)									
Destination: 0c:6d:5d:48:00:00 (0c:6d:5d:48:00:00)								0000	0c 6d 5d
.....0..... = LG bit: Globally unique address (factory default)								0010	00 7c 16
.....0..... = IG bit: Individual address (unicast)								0020	00 02 6d
Source: 0c:15:cd:16:00:01 (0c:15:cd:16:00:01)								0030	00 00 00
.....0..... = LG bit: Globally unique address (factory default)								0040	00 00 00
.....0..... = IG bit: Individual address (unicast)								0050	00 04 00
Type: IPv4 (0x0800)								0060	0e 0f 16
[Stream index: 0]								0070	1e 1f 20
Internet Protocol Version 4, Src: 10.0.0.1, Dst: 20.0.0.2								0080	2e 2f 30
0100 = Version: 4									
.... 0101 = Header Length: 20 bytes (5)									
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)									
Total Length: 124									
Identification: 0x10bd (4285)									
010. = Flags: 0x2, Don't fragment									
...0 0000 0000 0000 = Fragment Offset: 0									
Time to Live: 64									
Protocol: IPv6 (41)									
Header Checksum: 0x0b9a [validation disabled]									
[Header checksum status: Unverified]									
Source Address: 10.0.0.1									
Destination Address: 20.0.0.2									
[Stream index: 6]									
Internet Protocol Version 6, Src: 1000::a, Dst: 1002::a									
Internet Control Message Protocol v6									

Рис. 119: Анализ трафика в туннеле (пакет 2)

Инкапсуляция IPv6 в IPv4: Виден успешный обмен ICMPv6 ping-пакетами между 1002::a (PC2) и 1000::a (PC1)

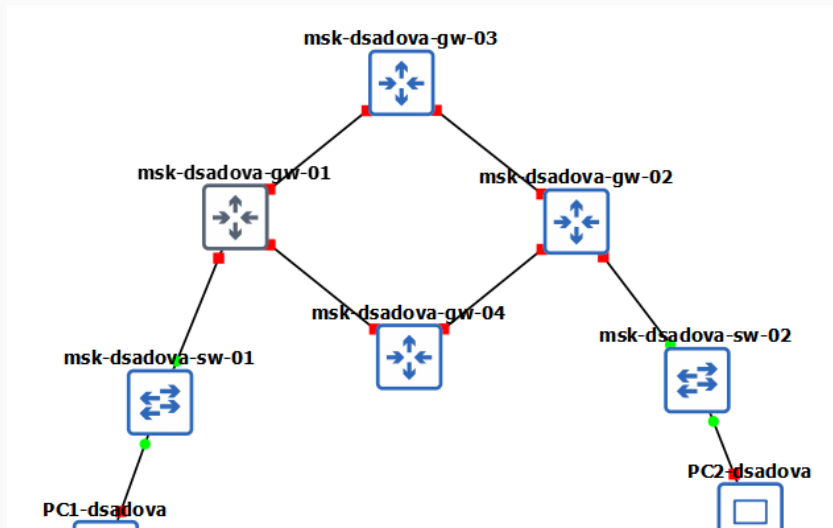
Протокол 41: В заголовке IPv4 указан протокол 41, что подтверждает инкапсуляцию IPv6

TTL/Нор Limit: - Внешний IPv4 заголовок: TTL=64 - Внутренний IPv6 заголовок: Нор Limit уменьшается с 63 до 61

Адреса туннеля: - Исходный: 10.0.0.1 (R1) - Конечный: 20.0.0.2 (R3)

Задание для самостоятельного выполнения

Заданы топология сети и адреса сегментов сети.



Требуется: – задокументировать моделируемую сеть: схемы L1, L3, заполнить таблицы адресации;

1. L1-схема (физический уровень)

Устройства: - PC1-user (VPCS) - PC2-user (VPCS) - msk-user-sw-01,
msk-user-sw-02 (L2-коммутаторы) - msk-user-gw-01 - msk-user-gw-02 -
msk-user-gw-03 - msk-user-gw-04

Соединения (L1):

Устройство 1	Интерфейс	Устройство 2	Интерфейс
PC1	eth0	sw-01	eth0
sw-01	eth1	gw-01	Gi0/0
gw-01	Gi0/1	gw-03	Gi0/0
gw-01	Gi0/2	gw-04	Gi0/0
gw-03	Gi0/1	gw-02	Gi0/0
gw-04	Gi0/1	gw-02	Gi0/1
gw-02	Gi0/2	sw-02	eth0
sw-02	eth1	PC2	eth0

2. Таблица сетей

Устройства	Сеть IPv4	Сеть IPv6
PC1-gw-01	10.10.1.96/27	2001:db8:1:1::/64
PC2-gw-02	10.10.1.64/28	2001:db8:1:6::/64
gw-01-gw-03	10.10.1.4/30	2001:db8:1:2::/64
gw-01-gw-04	10.10.1.8/30	2001:db8:1:3::/64
gw-03-gw-02	10.10.1.16/30	2001:db8:1:4::/64
gw-04-gw-02	10.10.1.32/30	2001:db8:1:5::/64

3. Таблица адресации интерфейсов

PC1

Параметр	Значение
IPv4	10.10.1.97/27
Шлюз	10.10.1.96
IPv6	2001:db8:1:1::10/64
Шлюз IPv6	2001:db8:1:1::1

PC2

Параметр	Значение
IPv4	10.10.1.65/28
Шлюз	10.10.1.64
IPv6	2001:db8:1:6::10/64
Шлюз IPv6	2001:db8:1:6::1

gw-01

Интерфейс	IPv4	IPv6
Gi0/0	10.10.1.96/27	2001:db8:1:1::1/64
Gi0/1	10.10.1.5/30	2001:db8:1:2::1/64
Gi0/2	10.10.1.9/30	2001:db8:1:3::1/64

gw-03

Интерфейс	IPv4	IPv6
Gi0/0	10.10.1.6/30	2001:db8:1:2::2/64
Gi0/1	10.10.1.17/30	2001:db8:1:4::1/64

gw-04

Интерфейс	IPv4	IPv6
Gi0/0	10.10.1.10/30	2001:db8:1:3::2/64
Gi0/1	10.10.1.33/30	2001:db8:1:5::1/64

gw-02

Интерфейс	IPv4	IPv6
Gi0/0	10.10.1.18/30	2001:db8:1:4::2/64
Gi0/1	10.10.1.34/30	2001:db8:1:5::2/64
Gi0/2	10.10.1.64/28	2001:db8:1:6::1/64

– настроить двойной стек IPv4 и IPv6, проверить соединения;

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-01
msk-dsadova-gw-01(config)# interface eth0
msk-dsadova-gw-01(config-if)# ip address 10.10.1.96/27
msk-dsadova-gw-01(config-if)# ip address 2001:db8:1:1::/64
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth1
msk-dsadova-gw-01(config-if)# ip address 10.10.1.5/30
msk-dsadova-gw-01(config-if)# ip address 2001:db8:1:2::1/64
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth0
msk-dsadova-gw-01(config-if)# ip address 10.10.1.96/27
msk-dsadova-gw-01(config-if)# ip address 2001:db8:1:1::1/64
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)# interface eth2
msk-dsadova-gw-01(config-if)# ip address 10.10.1.9/30
msk-dsadova-gw-01(config-if)# ip address 2001:db8:1:3::1/64
msk-dsadova-gw-01(config-if)# no shutdown
msk-dsadova-gw-01(config-if)# exit
msk-dsadova-gw-01(config)#
```

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-04
msk-dsadova-gw-04(config)# interface eth0
msk-dsadova-gw-04(config-if)# ip address 10.10.1.10/30
msk-dsadova-gw-04(config-if)# ip address 2001:db8:1:3::2/64
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)# interface eth1
msk-dsadova-gw-04(config-if)# 10.10.1.33/30
% Unknown command: 10.10.1.33/30
msk-dsadova-gw-04(config-if)# ip address 10.10.1.33/30
msk-dsadova-gw-04(config-if)# ip address 2001:db8:1:5::1/64
msk-dsadova-gw-04(config-if)# no shutdown
msk-dsadova-gw-04(config-if)# exit
msk-dsadova-gw-04(config)#
```

Рис. 122: Настройка IPv4 и IPv6 на устройствах

```
frr# configure terminal
frr(config)# hostname msk-dsadova-gw-02
msk-dsadova-gw-02(config)# interface eth0
msk-dsadova-gw-02(config-if)# ip address 10.10.1.64/28
msk-dsadova-gw-02(config-if)# ip address 2001:db8:1:6::1/64
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# interface eth1
msk-dsadova-gw-02(config-if)# 10.10.1.18/30
% Unknown command: 10.10.1.18/30
msk-dsadova-gw-02(config-if)# ip address 10.10.1.18/30
msk-dsadova-gw-02(config-if)# ip address 2001:db8:1:4::/64
msk-dsadova-gw-02(config-if)# ip address 2001:db8:1:4::2/64
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# interface eth2
msk-dsadova-gw-02(config-if)# ip address 10.10.1.34/30
msk-dsadova-gw-02(config-if)# ip address 2001:db8:1:5::2/64
msk-dsadova-gw-02(config-if)# no shutdown
msk-dsadova-gw-02(config-if)# exit
msk-dsadova-gw-02(config)# █
```

Рис. 123: Настройка IPv4 и IPv6 на устройствах

– настроить динамическую маршрутизацию сетей IPv4 и IPv6 для протоколов RIP, OSPF, проверить соединения и маршруты

- Изучили принципы маршрутизации в IPv4- и IPv6-сетях и принципы настройки сетевого оборудования.